End-TermExamination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B. Tech. IT/MAE/DMAM>, Semester:1st
(Feb-March, 2023)

Subject Code:BMA-110	Subject: Engineering Mechanics
Time :3 Hours	Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 Attempt any 8 questions **(2.5*8=20)**

(a) Define the system of forces.
(b) Discuss the cone of friction and coefficient of friction.
(c) Define the Moment of inertia and Radius of gyration.
(d) Explain the perfect, deficient and redundant frame.
(e) Discuss the normal and tangential component of velocity and acceleration in curvilinear motion.
(f) Define the coefficient of restitution. What will be the value of coefficient of restitution for elastic and inelastic bodies.
(g) A bar is sliding against wall at one end and the floor at other end. How these act can be split into motion of translation and rotation.
(h) State the work energy principle during the general plane motion.
(i) what do understand by the term Instantaneous centre of rotation
(j) State D'Alembert's principle

UNIT-I

Q2 Three parallel forces F1, F2, F3 are acting on a body as shown in fig and the body is in equilibrium. If force F1 = 250N and F3=1000N and the distance between F1 and F2=1.0m, then determine the magnitude of force F2 and the distance of F2 from force F3. **(10)**

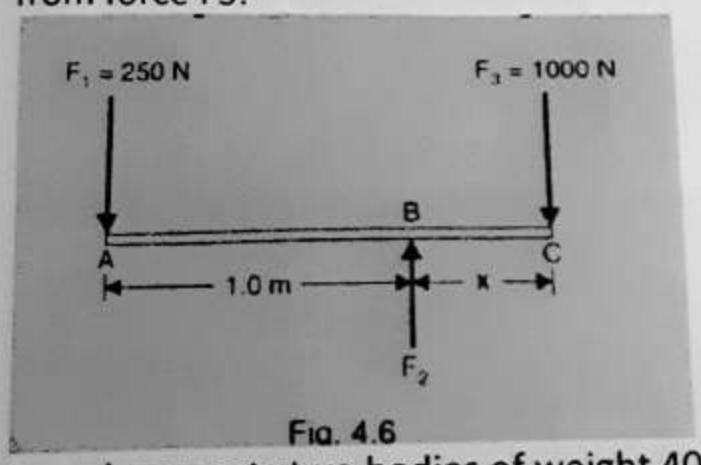
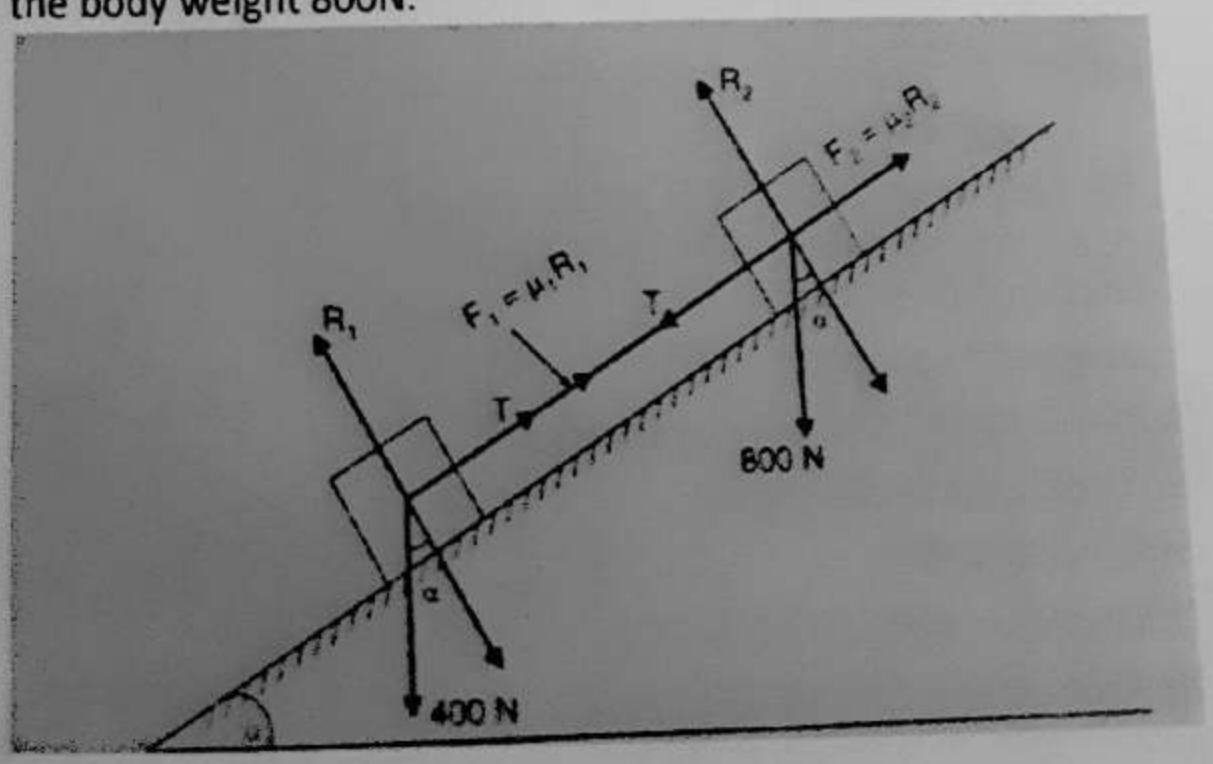


Fig. 4.6

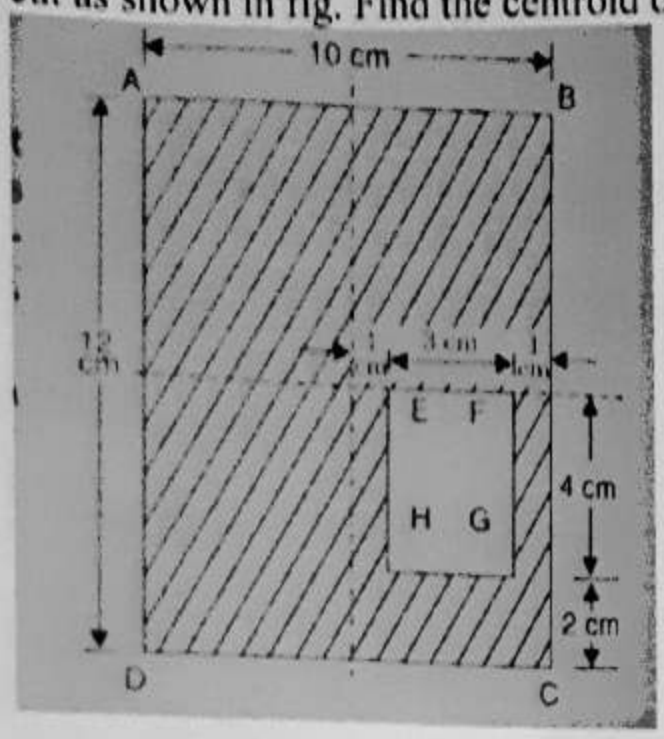
Q3 A cord connects two bodies of weight 400N and 800N. The two bodies are placed on an inclined plane and cord is parallel to inclined plane. The coefficients of friction for the weight of 400N is 0.15 and for 800N is 0.4. Determine the inclination of the plane to the horizontal and the tension in the cord when the motion is about to take place down the incline plane. Body weight 400N is below the body weight 800N. **(10)**



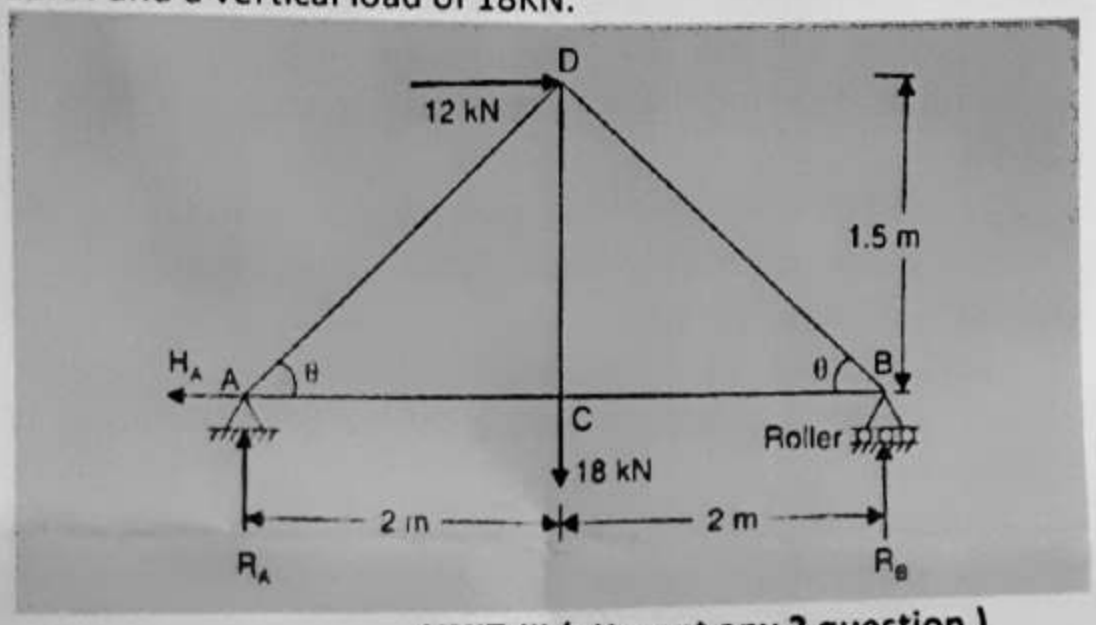
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UNIT-II

Q4 From a rectangular lamina ABCD 10cm *12cm a rectangular hole of 3cm*4cm is cut as shown in fig. Find the centroid of area. **(10)**



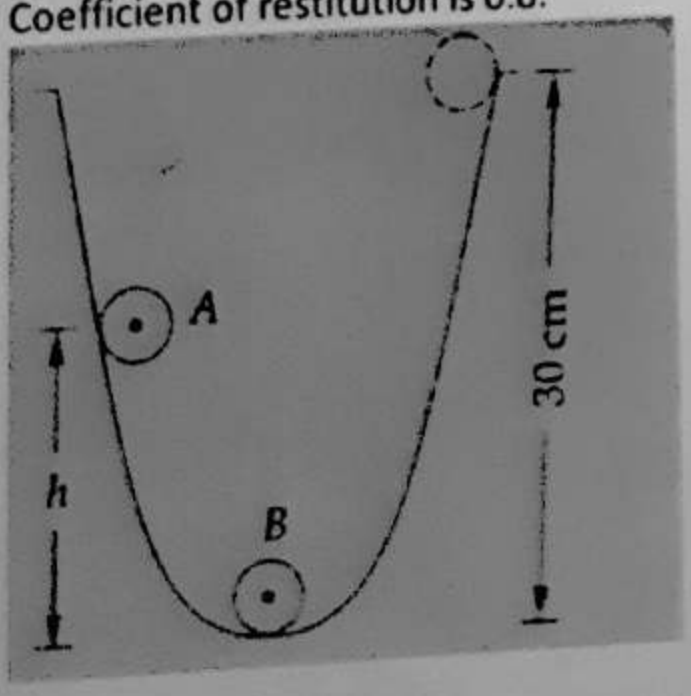
Q5 Determine the forces in the truss shown in fig which carries a horizontal load of 12KN and a vertical load of 18KN. **(10)**



UNIT-III (attempt any 2 question)

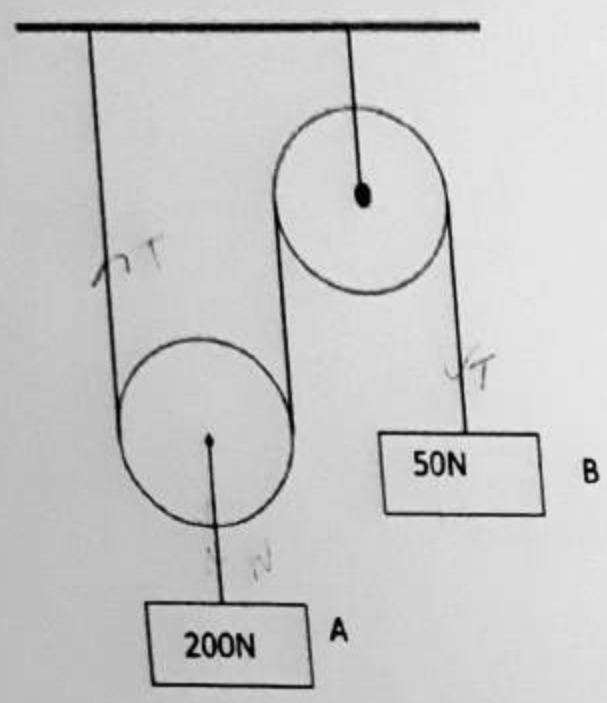
Q6 The initial angular velocity of a rotating body is 2rad/s and initial angular acceleration is zero. The rotation of the body is according to the relation $\alpha=3t^2-3$. Find i) angular velocity and ii) angular displacement when t=5sec. Consider angular displacement in radians and time in seconds. **(10)**

Q7a A spherical ball A of 100gm mass, when released from rest, slides down the surface of a smooth container of height 30cm. It collides with a spherical ball B of mass 25gm resting at the bottom of the container as shown in fig. What should be the height h from which the ball A be released so that after the impact the ball B just leaves the container. Coefficient of restitution is 0.8. **(10)**



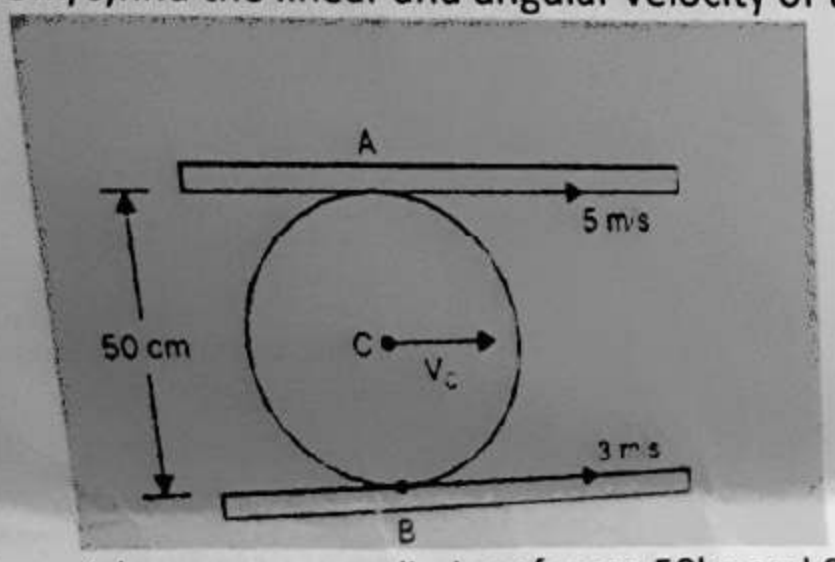
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Q7b Find the tension in the string and acceleration of blocks A and B weighing 200N and 50 N respectively, connected by a string and frictionless and weightless pulleys as shown in Fig below **10**



UNIT-IV

Q8 A cylindrical roller, 50cm in diameter is in contact with two conveyors at its top and bottom as shown in fig. If the belts run at the uniform speed of 5m/sand 3m/s,find the linear and angular velocity of the roller. **(10)**



Q9 A homogenous cylinder of mass 50kg and 0.4m in radius is having initial velocity of 5m/s down the 30° inclined plane. Calculate the velocity of cylinder when it has reached 10m down the plane from starting point. Assume the cylinder is rolling without sliding. **(10)**

